TEXORA · LECTURE SHEET

Work Study and Method Study

The systematic examination of how work is done — method study's SREDIM procedure for designing a better method, its recording and examination techniques, the precautions that make change stick, and how method study hands over to work measurement.

TE-4107 · PRODUCTION PLANNING AND CONTROL · TOPIC 2 · LEVEL 4 TERM 1

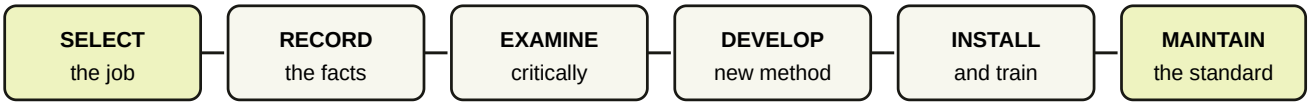
1. What work study is

Work study is the systematic investigation of all factors affecting the efficiency of a work situation, in order to achieve improvement. It has two inseparable halves. **Method study** asks: *is this the right way to do the job at all?* — it records and critically examines existing and proposed ways of working and develops easier, more effective methods. **Work measurement** asks: *how long should the job take when done by the approved method?* — its subject is Topic 3. Neither half works alone: timing a bad method merely standardises waste, while designing a good method without standard times leaves planning with nothing to schedule against.

The purpose of work study in a textile mill is concrete: raise productivity without capital investment, reduce operator fatigue, standardise best practice across shifts, provide the standard times on which costing, line balancing, incentives and delivery promises all depend, and expose hidden losses — waiting, double handling, unnecessary transport — that no production report shows.

2. Method study: the SREDIM procedure

Method study follows a fixed six-step procedure remembered as **SREDIM**:



Skipping EXAMINE produces decoration, not improvement; skipping MAINTAIN produces a method that lasts a fortnight.

Fig. 1 — The SREDIM procedure of method study.

1. **Select** the work to be studied. Choose by economic importance (high-cost, high-volume, bottleneck operations), technical signals (high rejection, accidents, complaints) and human considerations

(fatiguing, unpopular jobs). A study of a trivial operation wastes the technique's credibility.

2. **Record** all facts of the present method — by direct observation, using the recording techniques of Section 3 — completely and without interpretation. A record that silently "tidies up" the method destroys the study.
3. **Examine** the record critically with the questioning technique of Section 4: purpose, place, sequence, person, means — each challenged, each answer chased to alternatives.
4. **Develop** the improved method: eliminate, combine, rearrange, simplify. Evaluate candidate methods on cost, quality, safety and acceptance before choosing.
5. **Install** the new method: gain approval, train operators and supervisors, provide the new layout/aids, run it in with close support. Installation is a change-management exercise, not a memo.
6. **Maintain**: audit periodically that the installed method is actually being used; methods drift back unless standards, training and supervision hold them.

3. Recording techniques

TECHNIQUE	WHAT IT RECORDS	TYPICAL TEXTILE USE
Outline process chart	Only operations and inspections, in sequence	Whole route of a fabric from grey store to dispatch
Flow process chart (man / material / equipment types)	Operations, transports, inspections, delays, storages with distances and times	Following a dye-lot through the wet-processing plant
Two-handed process chart	Left- and right-hand activities simultaneously	Seamstress's motions at a sewing workstation
Multiple activity chart	Several workers/machines on a common time scale	Weaver and looms; operator and two knitting machines
Flow / string diagram	Path of movement on a scaled layout	Material flow in a cutting room; cone transport in winding
Travel chart	Frequency of movements between departments (from–to matrix)	Planning a new layout for the finishing plant

The five flow-process-chart symbols are standard: ○ **operation** (work done), ⇒ **transport** (movement), □ **inspection** (checking), **D delay** (waiting), ▽ **storage** (controlled holding). Only operations and inspections add value; the other three are cost to be minimised — the chart makes their share visible at a glance.

4. The critical examination technique

For every activity the examiner asks, in order: **Purpose** — what is done, why is it necessary, what else could achieve it, what should be done? **Place** — where is it done, why there, where else, where should it be? **Sequence** — when, why then, when else, when should it be? **Person** — who does it, why that person, who else, who should? **Means** — how is it done, why that way, how else, how should it be? The primary questions establish the facts; the secondary questions force alternatives. The output of examination is applied through four verbs: **Eliminate** the unnecessary, **Combine** what can be done together, **Rearrange** the sequence, **Simplify** what remains.

> **MINI-EXAMPLE — EXAMINATION IN A DYE HOUSE**
>
> A flow process chart of a dye-lot shows the lot is transported 46 m from the dyeing machine to a hydro-extractor, waits (D) an average of 35 min, is extracted, then travels back 30 m past the same machine to the dryer. **Sequence/place** questioning: the extractor stands where it was installed a decade ago for a different line. Rearranging — moving one extractor beside the dyeing row — removes both transports and most of the delay; combining extraction with the unloading crew's idle window removes the rest. No machine was bought; the method was the machine.

5. Precautions when introducing new methods

- **Involve the people affected from the start.** Operators who are studied without explanation assume the worst — usually that earnings or jobs are threatened — and the study fails socially before it fails technically. State the purpose honestly: method improvement, not speed-up by stealth.
- **Never build a standard on an unrepresentative trial:** a new method timed during its first enthusiastic week will not hold; allow learning to settle before work measurement fixes times.
- **Protect earnings during change**: where incentive schemes exist, guarantee that no operator loses pay because a method changed — otherwise every future study will be resisted, rationally.
- **Train, then supervise, then audit.** A method installed without training exists only on paper; supervisors must be trained first, since they will defend or quietly kill the method.
- **Watch quality and safety during transition**: output pressure during run-in raises rejections and accident risk; monitor both explicitly.
- **Document the standard method** (operations, layout, aids, settings) so that "the method" survives staff changes — an undocumented method is a rumour.

6. Relationship with work measurement

Method study and work measurement form one technique with a fixed order: **method first, measurement second**. Method study defines *what* the job is — the standard method — and work measurement then determines the *standard time* for that method (Topic 3). The standard time in turn feeds costing, production planning, line balancing and incentives; if any of those uses reveals the method has drifted, method study is re-applied. There is also a diagnostic loop: work-measurement data (activity sampling, Topic 4) often reveals where method losses hide, telling method study where to look next. Separately used, each technique is half blind: measurement without method freezes waste into standards; method without measurement produces improvements nobody can plan with.

Glossary

Work study — systematic investigation of work to improve efficiency; comprises method study and work measurement. **Method study** — recording and critical examination of ways of working to develop better methods. **SREDIM** — Select, Record, Examine, Develop, Install, Maintain. **Flow process chart** — chart using the five activity symbols to record a process in sequence. **Critical examination** — structured questioning of purpose, place, sequence, person and means. **Standard method** — the approved, documented way a job is to be performed.

Key points

• Work study = method study (better way) + work measurement (standard time); method always precedes measurement.
• SREDIM: Select, Record, Examine, Develop, Install, Maintain — no step is optional.
• Only operations and inspections add value; transport, delay and storage are the targets.
• Critical examination asks purpose, place, sequence, person, means — then eliminate, combine, rearrange, simplify.
• New methods fail socially before technically: involve operators, protect earnings, train, audit.

Practice

1. Put SREDIM in order and state, for a study of a garment finishing section, one concrete action under each step.
2. A flow process chart of a process totals 22 activities: 9 operations, 3 inspections, 6 transports, 3 delays, 1 storage. What percentage of activities adds value, and what does the chart tell the method engineer to attack first?
3. Why must method study precede work measurement? Answer in terms of what a standard time means.
4. List four precautions when installing a new method on a sewing floor and justify each in one line.

Answer key

1. Select → Record → Examine → Develop → Install → Maintain. Example actions: Select — choose the pressing/packing area because dispatch delays trace there; Record — flow process chart of a garment bundle from final inspection to carton; Examine — question why bundles travel to a central table and wait; Develop — move a pressing station into the line end and combine final inspection with thread trimming; Install — train the finishers, run the new layout for a trial week with support; Maintain — monthly audit with the section supervisor that the layout and sequence are still as installed.
2. Value-adding = operations + inspections = 12 of 22 ≈ 54.5 %. Nearly half the activities are transport, delay and storage. Attack the largest non-value block first — here transports (6) and delays (3): examine place and sequence to remove movement and waiting before refining the operations themselves.
3. A standard time is the time for a qualified worker to perform a *specified method* at standard performance. If the method is not fixed first, the time study measures whatever method the operator happened to use — waste included — and the "standard" then legitimises that waste in costing, planning and incentives, and differs from operator to operator. Method study fixes the one best way; measurement then times that way.
4. (i) Inform and involve operators early — prevents fear-driven resistance and surfaces practical objections. (ii) Protect earnings during the change — removes the rational incentive to defeat the study. (iii) Train operators and supervisors before go-live — an untrained method exists only on paper. (iv) Monitor quality and safety during run-in — transition pressure raises rejections and accidents if unwatched. (Also acceptable: document the standard method; audit after installation.)

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib